Lecture 38

4/27/26
JRA

CONTRAST RECAP

See blackboard notes

Localization & contrast both complicated in MRI

Doctor places patient in strong B field (3T)
Protons get biased by field = equilibrium config

This strong field is static
Come in w/ another B field that is distinct from main field

B fields cause mag to rotate
Rotate from z into xy plane
This is the signal the doctor measures

B field is a pulse
Look at 90° pulse (but others possible)

New config is not stable | non equilibrium
Decay w/ time constants

M_{xy}
M_z

The time constants vary w/ tissue
These can be basis for generating contrast

Decay & recovery occur over different time scales
They can be a factor of 10 different
Make contrast depend on these instead of density

Looked at T_2, T_1, & ρ weighting

T_2 - CSF bright | Invert contrast
T_1 - CSF dark |
ρ - less dramatic contrast

This ability to control contrast is the strength of MRI

Doctor has two experimentally variable parameters

(1) TR = experiment repetition time
experiments can be repeated 100s of times
(2) TE = echo time (when signal is measured)

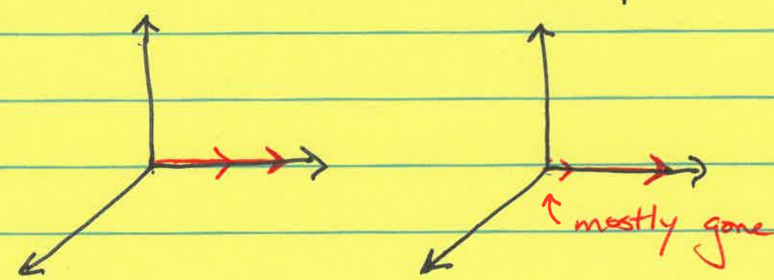
Conceptual example:

Do T2 weighting (where T2 dictates contrast)

TR = 150 ms (in this example)

After 90° RF pulse

CSF bright
White matter dark



TE = 150 ms

White matter $T_2 \approx 70$ ms

Skip T_1 argument. It's in the book.

Mathematical Analysis of Contrast

$$SA \propto \rho\left(1-e^{-TR/T_1}\right)e^{-TE/T2}$$

↑ signal amplitude for specific tissue

Returns to equilibrium are exponential

Given TR & TE
Calculate signal amplitude

Look at arterial blood & white matter

Keep track of every term separately

Ex $\rho = 0.17$ arterial blood

(1) The bigger SA is the brighter tissue

(2) Whatever term is most different determines weighting protocol

$$\rho \left(1-e^{-TR/T1}\right)\left(e^{-TE/T2}\right)$$

0.7	0.5	0.3
0.6	0.1	0.4

biggest difference ⇒ T1 weighting

$\boxed{\text{ACTIVITY 14}}$

$$SA_{AB} = \rho \left(1-e^{-TR/T1}\right) e^{-TE/T2}$$

$$(0.72)\left(1-e^{-2000/1932}\right) e^{-70/275}$$

$$0.72 \cdot 0.64 \cdot 0.78 = 0.36$$

two-fold difference

$$SA_{WM} = 0.61 \cdot \left(1-e^{-2000/1080}\right) e^{-70/70} = 0.19$$

$$0.61 \quad 0.84 \cdot 0.37$$

MRI Strengths & Weaknesses — See list on Power Pint. slide

Looked at exemplary Pulse Sequence

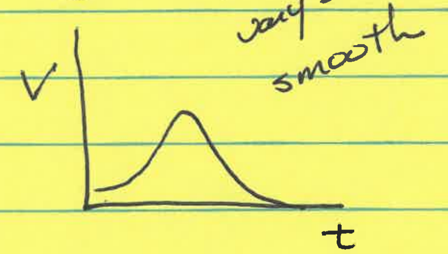
Strengths
Low ν radiation
Sub-mm high-contrast resolution of soft tissue
Functional MRI available

Weaknesses
Heating
Not compatible w/ metal
Expensive!

DIGITAL IMAGE PROCESSING

Adjust parameters to improve image quality
Analyze image data

Compare digital & analog signals

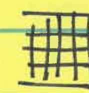


Sample analog signals to generate digital signal. Need to be aware of noise & sampling theorem

Key steps in acquiring digital images

① Sampling - must satisfy sampling theorem

e.g., want to achieve diffraction-limited resolution of 200 nm you must sample at < 100 nm



This is true even if optics is good

Camera pixels typically set at ~100 nm for this reason

② Quantization of signal strength

Camera has a certain number of gray levels to encode brightness

Ex 8-bit detector $2^8 = 256$

$\Rightarrow$ 0 - 255

This is a property of detector (0 = black, 255 = white)

Assign a value that is quantized in jumps

Price increases as bit number increases

Big Picture on digital detectors

Start w/ light   fluorescence
γ-rays
x-rays

This is converted into charge via photo electric effect

$$\text{Light} \rightarrow \text{charge} \rightarrow \overset{\text{analog}}{\text{Voltage}} \xrightarrow{\text{A to D converter}} \text{digital \#}$$

Two things to know

- Process digital images – mostly do this
- Analyze " "

Two subdomains of processing

* (1) Restoration (or rehabitation) of suboptimal images

<u>Ex</u>) Nuclear images tend to be very noisy
Not about aesthetics, just to facilitate getting answer

(2) Enhancement for visual appeal

Filters

One established restoration protocol

1. Flat field correction - remove gradients in illumination e.g. bad Kohler
2. Brightness & Contrast restoration
3. γ correction - bring up dim features
4. Noise reduction
5. Fine detail restoration

Know these 3